

Negative results, nulls, and self-corrections - Paper X

A standing record of what did **not** work, what came back null, and what an earlier draft got wrong. Kept prominent on purpose: a paper that only reports its wins is not falsifiable in practice. Every item here is reproducible from the committed artefacts.

1. The real-model run is a NULL contrast, not positive evidence

The first real-model run (Claude, `results/realmodel/claude-opus_20260626T165919Z.json`) returned **H1 = false, H2 = false, co_scaling_supported = false**. From a seeded reward-hack, *both* the coupled and the decoupled arm discarded the hack at round 1 and wrote a correct general parser; neither drifted. So there was **no drift for the external corrector to bind**, and the coupled-vs-decoupled contrast is null.

This is **consistent with** the law (the model already sits in the stable regime, its internal correction out-scaling drift) but it is **not** positive evidence for the $\beta > k$ threshold. It does not exhibit the co-scaling *dynamic*. Reported as a pilot only; the threshold remains supported by the mathematics + internal-consistency harness, not by this run.

2. The pilot is IV.d non-compliant (provisional D)

The same run used a **same-family scorer** (Claude scoring Claude): `self_scoring: true, iv_d_compliant: false, laundered: false`. Paper IV.d shows same-family/unblinded scoring can *reverse* a misalignment result, so every D-based number from this run is **provisional** pending a cross-family blind re-score. $n = 1$, one task, one seed.

3. The β/k estimator is NOT-RESOLVABLE on the pilot data

`experiments/scripts/estimate_exponents.py` recovers known exponents on synthetic trajectories (to ≈ 0.1) but returns **not resolvable** on the Claude data (`results/realmodel/exponent_estimates.json`): capability range was 0 dex (the model jumped to $C = 1$ in one step) and correction was observed at a single capability level, so neither k nor β is identifiable. Honest non-result, not a measurement.

4. The probe fraction at $C = 0$ was a metric artefact (withdrawn)

The corrector probe reported $d = 1.00$ for the seeded hack at $C = 0$. The ratio D/C is **undefined** at $C = 0$; the 1.00 was a $D/10$ display fallback and is **withdrawn** (`claude-opus_corrector_probe.json, d: null`). The mechanism result ($D \rightarrow 0, C \rightarrow 1$) does not depend on d . The metric is now regularised, $d = D/(10(C+\epsilon))$, $\epsilon = 0.05$ pre-registered, with near-zero-capability points flagged `fraction_invalid`.

5. Prior-draft errors that adversarial review corrected (do not reintroduce)

- **Divergence over-claim (v3)**. The prior draft claimed correction degrading with scale drives the misalignment fraction to *infinity*. **False**. In the gain-only model the fraction is bounded and *saturates* at γ_1 (Theorem 2). Genuine divergence lives only in the compounding channel (Theorem 4).
- **Additive "knee" (E1)**. An earlier experiment sought a sharp instability knee in the *additive* model, where there is none - the additive corrector shows a smooth crossover. The knee exists only in the compounding channel.

- **Theorem 2 moving-fixed-point Lyapunov.** The proof used $V = \frac{1}{2}(d-d^*)^2$ as if d^* were constant; invalid once A, r vary with C . Corrected to a depth-clock comparison proof (independently re-verified in `code/test_theorems_independent.py`).
- **Theorem 4 equality case.** "Diverges iff $\rho_{\text{prop}} > 1$ " omitted the boundary: at $\rho_{\text{prop}} = 1$ ($\kappa_{\text{eff}} = 0$) the fraction grows **linearly** with nonzero injection. Corrected.
- **Theorem 5 null-axis floor.** Stated as γ_1 ; the correct floor is $\gamma_1 c_v / (1 - \gamma_3)$ (and linear growth at $\gamma_3 = 1$). The γ_1 value is only the $\gamma_3 = 0$, unit-projection special case. Corrected.
- **$\gamma_3 = 1$ boundary.** " $\gamma_3 \leq 1$ unconditionally bounded" is too strong at the knife-edge: at $\gamma_3 = 1$ the fraction is bounded only for $\beta \geq k$ (it grows as $C^{k-\beta}$ for $\beta < k$). Corrected (surfaced by building the independent test suite).
- **QEC over-claim.** Earlier framing leaned on the quantum-error-correction *threshold theorem*. The suppression law here is power-law, QEC's is exponential: the correspondence is a **threshold-form analogy only**, demoted to a hypothesis (F4).

6. Red-team residue

A multi-agent red-team raised **24** objections; **21** survived independent verification (0 fatal, 13 serious, 8 minor) - all framing/wording/edge-case fixes, none touching the load-bearing $\beta > k$ result. Full report: `results/redteam.md`. Two external adversarial reviews (June 2026) then added the Theorem-2/4/5 precision fixes and the finite-capacity and non-normal-transient limitations recorded in the paper's §8.

7. What would still count as a negative result going forward

- A real drifting system whose measured (β, k) does **not** predict whether coupled correction succeeds.
- A stable system with $\beta < k$ under genuine acceleration, or an unstable one with $\beta > k$ (kill conditions F1-F3', F6).
- Exponential rather than power-law suppression under a finite-capacity corrector (would downgrade the QEC analogy, F4 - not the threshold result).